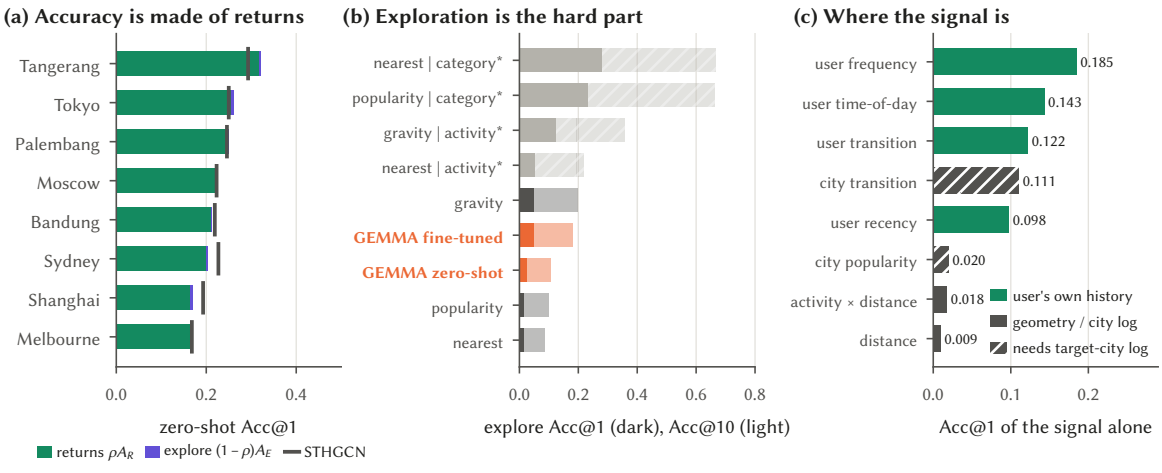
(a) Accuracy is made of returns
Tangerang
Tokyo
Palembang
Moscow
Bandung
Sydney
Shanghai
Melbourne
0.0
0.2
0.4
zero-shot Acc@1
returns ρA_R
explore $(1-\rho)A_E$
STHGCN
(b) Exploration is the hard part
nearest | category*
popularity | category*
gravity | activity*
nearest | activity*
gravity
GEMMA fine-tuned
GEMMA zero-shot
popularity
nearest
0.0
0.2
0.4
0.6
0.8
explore Acc@1 (dark), Acc@10 (light)
(c) Where the signal is
user frequency
user time-of-day
user transition
city transition
user recency
city popularity
activity × distance
distance
0.185
0.143
0.122
0.111
0.098
0.020
0.018
0.009
user's own history
geometry / city log
needs target-city log
0.0
0.1
0.2
Acc@1 of the signal alone